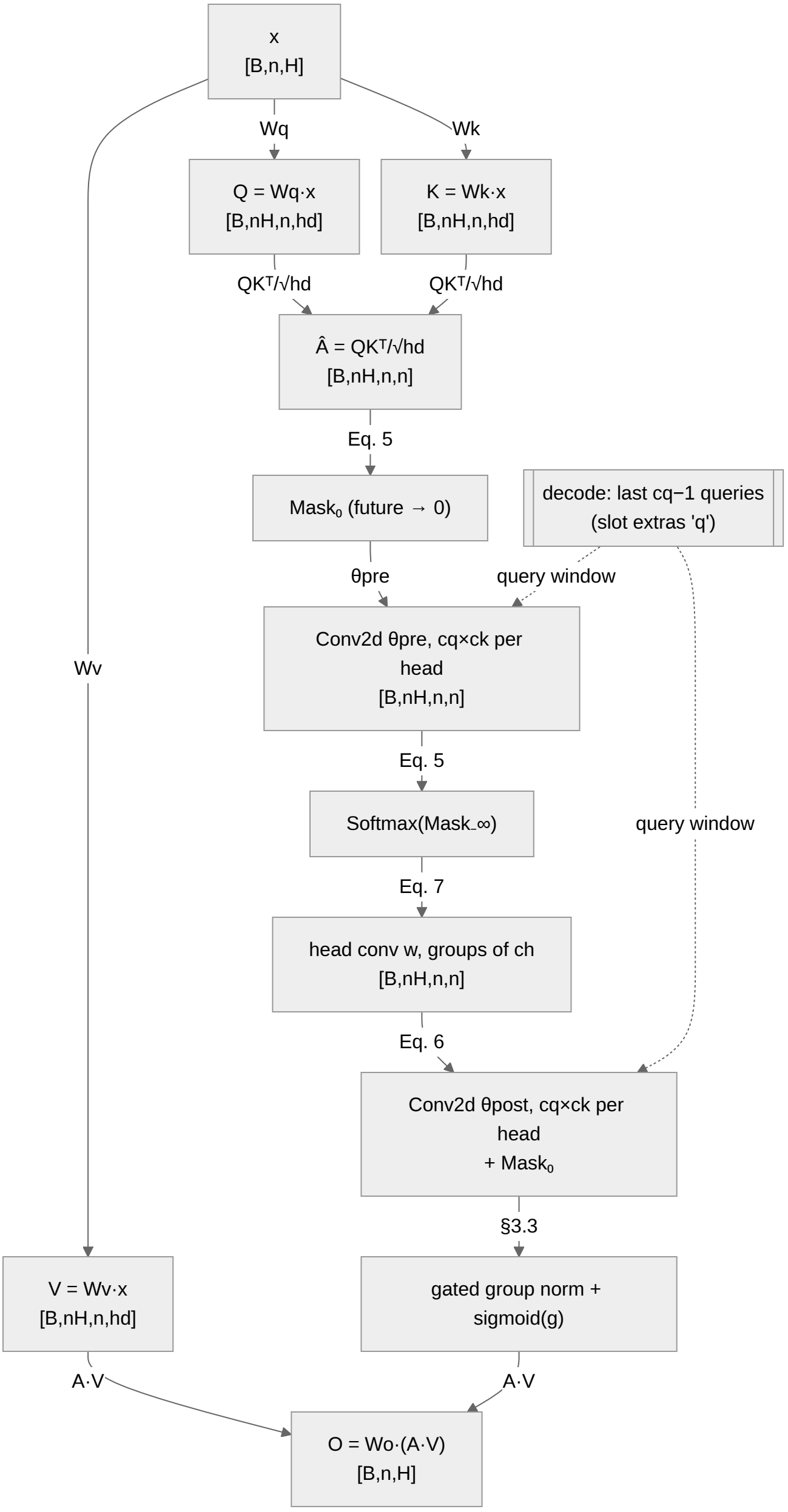
x
[B,n,H]
Wq
Wk
Wv
Q = Wq·x
[B,nH,n,hd]
K = Wk·x
[B,nH,n,hd]
V = Wv·x
[B,nH,n,hd]
QKᵀ/√hd
QKᵀ/√hd
Â = QKᵀ/√hd
[B,nH,n,n]
Eq. 5
Mask₀ (future → 0)
θpre
decode: last cq−1 queries
(slot extras 'q')
query window
Conv2d θpre, cq×ck per
head
[B,nH,n,n]
Eq. 5
Softmax(Mask_∞)
Eq. 7
head conv w, groups of ch
[B,nH,n,n]
Eq. 6
query window
Conv2d θpost, cq×ck per
head
+ Mask₀
§3.3
gated group norm +
sigmoid(g)
A·V
A·V
O = Wo·(A·V)
[B,n,H]